

Kathryn A. Mummah, PhD

NUCLEAR ENGINEER SPECIALIZING IN FUEL CYCLE MODELING AND SAFEGUARDS

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EDUCATION

University of Wisconsin–Madison

Madison, WI

PhD Nuclear Engineering & Engineering Physics

May 2025

MS Nuclear Engineering and Engineering Physics

May 2020

- Advisor: Professor Paul P.H. Wilson
- Dissertation: *State-Level Nuclear Fuel Cycle Simulations for International Safeguards Applications*
- Graduate Certificates in Science Communication (LSC) and Energy Analysis & Policy (EAP)

University of Illinois at Urbana-Champaign

Urbana, IL

B.S. Nuclear, Plasma, and Radiological Engineering, graduated with Honors, GPA: 3.6 / 4.0

May 2017

- Concentration in Power, Safety, and the Environment
- Minor in Atmospheric Sciences

EXPERIENCE

Los Alamos National Laboratory

Los Alamos, NM

R&D Engineer, NEN-5 Nuclear Systems

Mar. 2025 - **PRESENT**

Los Alamos National Laboratory

Los Alamos, NM

Graduate Research Assistant, NEN-1 Safeguards Science & Technology

Jan. 2021 - 2024

- Built tool to convert cradle-to-grave nuclear fuel cycle simulations into IAEA-style nuclear material accounting reports.
- Generated realistic synthetic country-level nuclear fuel cycle simulations and characterized system-level behavior
- Advanced agent-based modeling at the material balance area level with high fidelity representations of nuclear material movements

UW-Madison Department of Engineering Physics

Madison, WI

Graduate Research Assistant, Computational Nuclear Engineering Research Group (CNERG)

Sept. 2017 - Feb. 2025

- Integrated acquisition pathway analysis into nuclear fuel cycle simulations
- Reviewed public communication efforts in consent-based siting of radioactive waste disposal facilities worldwide
- Designed uranium enrichment cascades in the Cyclus fuel cycle code to model post-JCPOA cascade performance scenarios

Grader

- NE405: Nuclear Reactor Theory, Spring 2018

Los Alamos National Laboratory

Los Alamos, NM

Grad Research Assistant, NEN-2 Advanced Nuclear Technology & Glenn T. Seaborg Summer Research Fellow

May 2018 - Aug. 2018

- Identified approaches to combine fuel cycle simulators and existing methodology for analyzing nuclear material diversion pathways.
- Taught nuclear engineering primer series including detailed background on nuclear energy industry and accidents

Visiting Student Guest, NEN-5 Systems Design & Analysis

May 2017 - Aug. 2017

- Developed MCNP decks for nuclear thermal rockets
- Created a reactor lab module to incorporate nonproliferation concepts into undergraduate curricula

University of Illinois Department of Nuclear, Plasma, and Radiological Engineering

Urbana, IL

Undergrad Research Assistant, Advanced Reactors and Fuel Cycle Research Group

Aug. 2016 - May 2017

Undergrad Research Assistant, Analysis of Reactor Transients and Stability Research Group

Jan. 2016 - Dec. 2016

Undergrad Research Assistant, Experimental Thermal Hydraulics Research Group

Aug. 2015 - Dec. 2015

Teaching Assistant

- Lead Engineering Learning Assistant, ENG 100 for Nuclear, Plasma, and Radiological Engineering. Fall 2015 & '16
- NPRE 100: Introduction to Nuclear, Plasma, and Radiological Engineering. Fall 2014, '15, & '16
- NPRE 101: Introduction to Energy Sources. Spring 2014 & '15

Grader

- ASTR 100: Introduction to Astronomy. Fall 2014

Idaho National Laboratory

Idaho Falls, ID

Intern, Fuels Modeling & Simulation

May 2016 - Aug. 2016

Oyster Creek Generating Station

Intern, Reactor Engineering

[Forked River, NJ](#)

Jun. 2015 - Aug. 2015

Exelon Generation

Intern, Spent Fuel & Decommissioning

[Warrenville, IL](#)

Jun. 2014 - Aug. 2014

Fermi National Accelerator Laboratory

QuarkNet Summer Intern, Optical Astronomy

[Batavia, IL](#)

Jun. 2012 - Aug. 2012

HONORS & AWARDS

AWARDS

2025	Wisconsin Initiative for Science Literacy Award for Communicating PhD Research to the Public
2022	American Nuclear Society Presidential Citation
2022	Best Presentation in Nuclear Technology & Society, ANS 2022 Student Conference
2022	Semi-finalist, Fulbright U.S. Student Program
2020	First place, Institute of Nuclear Materials Management J.D. Williams Student Paper Award
2017	Daniel F. Hang Outstanding Senior Design Award
2017	University of Illinois Knights of St. Patrick
2016	Alpha Nu Sigma
2016	Best Presentation in Public Image, ANS 2016 Student Conference
2016	William R. Schowalter Award
2016	American Nuclear Society Commendation for Service and Leadership
2014	American Nuclear Society UIUC Most Committed Member
2013	University of Illinois James Scholar

SCHOLARSHIPS AND FELLOWSHIPS

2024-25	Graduate Fellow, Consortium for Monitoring, Technology, and Verification
2019-24	Graduate Fellow, DOE/NNSA Nuclear Nonproliferation International Safeguards Program (NNIS)
2022-23	ANS Robert A. Dannels Memorial Scholarship
2021-22	ANS Graduate Scholarship
2020-21	ANS Michael J. Lineberry Memorial Scholarship
2018	Summer Research Fellow, Glenn T. Seaborg Institute
2017-19	Graduate Fellow, Consortium for Verification Technology
2017-19	ANS Fuel Cycle and Waste Management Division Randall Scholar
2015-17	DOE Nuclear Energy University Program IUP Scholarship Award
2015-17	Nuclear Regulatory Commission Scholarship
2015-17	American Nuclear Society Decommissioning & Environmental Sciences Scholarship
2016-17	Roy G. Post Foundation Scholarship
2016-17	Dale W. and Wanda L. Weaver Engineering Scholarship
2016-17	Crowe Horwath Scholarship
2016-17	Edith and Harry Darby Leadership Scholarship
2015-17	Catherine Pritchard Undergraduate Scholarship
2013-17	Mike Harper Leadership Scholarship
2015-16	Exelon Energy for Education Scholarship Award
2014-16	National Academy for Nuclear Training Scholarship
2014-15	ANS Incoming Sophomore Scholarships

PAPERS AND PRESENTATIONS

CONFERENCES PAPERS AND ABSTRACTS

- [12] **Mummah, K. A.** and Wilson, P. P. **Jun. 2024**, "Cyclus toolkit enhancements to simulate nuclear material buying patterns," in *Transactions of the American Nuclear Society, Annual Meeting 2024*, vol. 130, Las Vegas, NV.
- [11] **Mummah, K. A.**, Jackson, D. E., Apt, K. E., Oakberg, J., Henzl, V., and Wilson, P. P. **May 2023**, "Simulated State Accounting Data for Development of Advanced and Automated State-level Safeguards Analysis," in *Proceedings of the 64th Annual Meeting of the Institute of Nuclear Materials Management*, Vienna, Austria.

- [10] Calice, M., Brossard, D., Edwards, M., Gruder, S., **Mummah, K. A.**, Nemet, G., Williams, S., and Wilson, P. P. **Nov. 2022**, “The Local Energy Transition: Perspectives on Community Energy Management in Wisconsin,” presented at the Behavior, Energy and Climate Change (BECC) Conference, Washington D.C.
- [9] **Mummah, K. A. Apr. 2022**, “Learning from the ”Successful Failure” of a Low- and Intermediate-Level Radioactive Waste Project as a Case Study in Consent-Based Siting,” presented at the 2022 American Nuclear Society Student Conference, Champaign, IL.
- [8] **Mummah, K. A.** and Wilson, P. P. **Jul. 2020**, “Integrating Acquisition Pathway Analysis Into The Cyclus Fuel Cycle Simulator,” in *Proceedings of the 61st Annual Meeting of the Institute of Nuclear Materials Management*, Palm Desert, CA, USA.
- [7] **Mummah, K. A. Jun. 2019**, “Material Diversion Analysis Within a Fuel Cycle Simulator,” presented at the 2019 Technical Workshop of Fuel Cycle Simulation, Champaign, IL.
- [6] **Mummah, K. A.**, Bahran, R., Miller, K., and Wilson, P. P. **Jul. 2019**, “Acquisition Pathway Analysis using Fuel Cycle Simulators,” in *Proceedings of the 60th Annual Meeting of the Institute of Nuclear Materials Management*, Palm Desert, CA, USA.
- [5] Mouginot, B., **Mummah, K. A.**, and WILSON, P. P. **Nov. 2018**, “Assessing Material Inventory Uncertainties in Integrated Fuel Cycle Simulations,” presented at the Advances in Nuclear Nonproliferation Technology and Policy Conference (ANTPC 2018), Orlando, FL.
- [4] Mouginot, B., **Mummah, K. A.**, and Wilson, P. P. **Jul. 2018**, “Evaluating computational methods for modeling off-normal operation of gas centrifuge cascades,” in *Proceedings of the 59th Annual Meeting of the Institute of Nuclear Materials Management*, Baltimore, MD, United States. [Online]. Available: https://resources.inmm.org/system/files/annual_meeting_proceedings/2018/a161_2.pdf.
- [3] Mouginot, B., **Mummah, K. A.**, and Wilson, P. P. **Jun. 2018**, “Modeling Potential JCPOA Diversion Scenarios with Cyclus,” presented at the 2018 American Nuclear Society Annual Meeting, Philadelphia, PA.
- [2] **Mummah, K. A.**, Bae, J. W., O’Grady, D., and Lopez, A. **Apr. 2017**, “NuWWIS: An Interim Storage Solution for Spent Fuel,” presented at the 2017 American Nuclear Society Student Conference, Pittsburgh, PA.
- [1] **Mummah, K. A.** and Kuprianczyk, C. **Apr. 2016**, “Public Image in Spent Fuel Disposal: Lessons Learned from Sweden’s SKB,” presented at the 2016 American Nuclear Society Student Conference, Madison, WI, United States.

TECHNICAL REPORTS

- [2] **Mummah, K. A.**, Jackson, D. E., Oakberg, J., Apt, K. E., and Henzl, V. **Sep. 2023**, “Advanced Algorithms for Scrutiny of Mandatory State Reports Declarations to the IAEA (Final Project Report),” Los Alamos National Laboratory (LANL), Los Alamos, NM (United States), Tech. Rep. LA-UR-23-32729. doi: 10.2172/2208761.
- [1] Henzl, V., **Mummah, K. A.**, and Apt, K. **2022**, “Analysis of States’ Cadence of Nuclear Operations: Detecting Safeguards Violations,” Los Alamos National Laboratory, Los Alamos, NM, Tech. Rep. LA-CP-22-20328.

POSTERS

- [4] **Mummah, K. A. Mar. 2021**, “Good Practices and Clear Communication Gaps in the Effort to Site a Low- and Intermediate-Level Deep Geologic Waste Repository in Ontario,” Poster, presented at the Waste Management Symposium (Virtual).
- [3] **Mummah, K. A.**, Mouginot, B., and Wilson, P. P. **Apr. 2018**, “Off-normal Operation in Gas Centrifuge Enrichment Cascades,” presented at the 2018 American Nuclear Society Student Conference Poster Session, Gainesville, FL, United States.
- [2] **Mummah, K. A.** and Williamson, R. **Nov. 2016**, “Investigating the Effects of Fuel Pellet Geometry on Pellet Cladding Mechanical Interaction using BISON,” presented at the 2016 American Nuclear Society Winter Meeting, Las Vegas, NV, United States.
- [1] **Mummah, K. A. Nov. 2015**, “Reactivity Management: More than Just Reactor Engineers,” presented at the 2015 American Nuclear Society Winter Meeting, Washington D.C., United States.

SPONSOR/CONSORTIUM MEETINGS AND WORKSHOPS

- [10] **Mummah, K. A.** and Wilson, P. P. **Feb. 2025**, “Improving inter-facility nuclear material movements in a fuel cycle simulator in support of synthetic nuclear material accounting,” presented at the Consortium for Monitoring, Technology, & Verification (MTV) Fuel Cycle Facility Modeling Workshop 2025.

- [9] **Mummah, K. A.** and Wilson, P. P. **Mar. 2024**, *Improved Time Behavior and Inventory Management to Support Nuclear Material Accounting in Cyclus*, Poster, Ann Arbor, MI.
- [8] **Mummah, K. A.**, Jackson, D. E., Apt, K., Oakberg, J., Henzl, V., and Wilson, P. P. **Mar. 2023**, *Progress in Simulated State Accounting Data*, Poster, Ann Arbor, MI.
- [7] **Mummah, K. A.**, Henzl, V., Apt, K., and Wilson, P. P. **Mar. 2022**, “Creating Synthetic State Declarations Using Cyclus,” presented at the 2022 Consortium for Monitoring, Technology, & Verification (MTV) Workshop.
- [6] **Mummah, K. A.** and Wilson, P. P. **Mar. 2021**, “Systems modeling of the nuclear fuel cycle to enhance safeguards implementation,” Poster, presented at the Consortium for Monitoring, Technology, and Verification Virtual Workshop 2021 (Virtual). [Online]. Available: <https://prezi.com/v/wineskjpbj6x/mtv2021/#>.
- [5] **Mummah, K. A.** and Wilson, P. P. **Oct. 2019**, “Can We Study Material Diversion Analysis Within a Fuel Cycle Simulator?,” presented at the Consortium for Monitoring, Technology, & Verification (MTV) Fuel Cycle Facility Modeling Workshop 2019, Madison, WI.
- [4] **Mummah, K. A.** and Wilson, P. P. **Jun. 2019**, “Extending the Cyclus Fuel Cycle Simulator for International Safeguards,” presented at the 2019 NNSA University Program Review, Raleigh, NC.
- [3] **Mummah, K. A.**, Mouginot, B., and Wilson, P. P. **Jun. 2018**, “Progress on Studying Gas Centrifuge Cascade Behavior of Off-Normal Enrichment Scenarios,” presented at the 2018 NNSA University Program Review.
- [2] **Mummah, K. A.** and Wilson, P. P. **Oct. 2018**, “Simulation of the Nuclear Fuel Cycle and its Applications to International Safeguards,” presented at the 2018 Consortium for Verification Technology (CVT) Workshop, Ann Arbor, MI, United States.
- [1] **Mummah, K. A.**, Mouginot, B., and Wilson, P. P. **Nov. 2017**, “Modeling Post-JCPOA Cascade Performance with Cyclus,” presented at the 2017 Consortium for Verification Technology Workshop, Ann Arbor, MI, United States.

SERVICE AND LEADERSHIP

PROFESSIONAL AFFILIATIONS AND SERVICE

American Nuclear Society (ANS)

Vice Chair, Decommissioning & Environmental Sciences Division
 Referee/Reviewer
 Student Conference Judge

June 2024 - **PRESENT**
 2019 - **PRESENT**
 2022 - **PRESENT**

Chair, Student Sections Committee (SSC)
 Treasurer, Decommissioning & Environmental Sciences Division
 Executive Committee, Fuel Cycle and Waste Management Division
 Committee Member, Student Sections Committee
 Chair, Liaison Subcommittee of the Student Sections Committee
 Executive Committee, Young Members Group
 Vice Chair, Student Sections Committee
 Executive Committee, Nuclear Nonproliferation Policy Division
 Committee Member, Diversity and Inclusion Committee
 Student Director, ANS Board of Directors
 Chair, Commendations Subcommittee of the Student Sections Committee
 Committee Member, Local Sections Committee

June 2021 - June 2024
 June 2022 - June 2024
 June 2021 - June 2024
 June 2016 - June 2024
 Apr. 2021 - June 2023
 June 2019 - June 2022
 June 2019 - June 2021
 June 2018 - June 2021
 July 2018 - June 2021
 June 2017 - June 2019
 June 2016 - Jan. 2021
 June 2016 - June 2019

ACADEMIC SERVICE

American Nuclear Society, University of Wisconsin-Madison Student Section

Member
 Governor
 Advocacy co-chair

Aug. 2017-May 2022
 May 2018 - Dec 2018
 Jun. 2017 - Nov. 2017

University of Wisconsin-Madison Department of Engineering Physics

Department Representative to the College of Engineering Graduate Advisory Committee
 Chair, Student Advisory Committee

Aug. 2018 - Nov. 2022
 Apr. 2019-Jan. 2022

Wisconsin Baja Racing

Steering Team Member

Aug. 2018 - May 2020

American Nuclear Society, University of Illinois Student Section

Member	Aug. 2013 - May 2017
President	Apr. 2015 - Apr. 2016
Internal Vice President	Apr. 2014 - Apr. 2015
Freshman President	Oct. 2013 - Apr. 2014

Women in Nuclear, University of Illinois Student Section

Cofounder	Aug. 2015
Member	Aug. 2015 - May 2017

University of Illinois Department of Nuclear, Plasma, and Radiological Engineering

Undergraduate Advisory Board Representative	Apr. 2014 - Dec. 2016
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Engineering Ambassadors, University of Illinois

Recruitment & Publicity Chair	Apr. 2016 - Dec. 2016
Ambassador	Nov. 2015 - May 2017

Engineering Council, University of Illinois

Student Introduction to Engineering (SITE) Reservations Chair	May 2015 - May 2016
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University of Illinois Student Chapter of the American Meteorological Society

Secretary	Apr. 2015 - Apr. 2016
Member	Jan. 2015 - May 2017

Illinois Space Society

RASC-AL Technical Team Member	Aug. 2014 - Dec. 2014
Representative to Students for the Exploration and Development of Space (SEDS)	Sept. 2013 - Sept. 2014

OTHER SERVICE

Generation Atomic

Advisory Board Member	July 2020 - Jan. 2023
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Rethinking Nuclear/Friends of Fission Midwest Chapter

Science Advisor	2018 - 2022
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Phi Mu - Delta Beta Chapter

Academic Excellence Chair	Dec. 2016 - May 2017
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INVITED TALKS

American Nuclear Society , ANS Members Take on Social Media Webinar, <i>Panelist</i>	July 29, 2024
Penn State Engineering Equity Summit/ANS Student , Radiant Women in Nuclear, <i>Moderator</i>	April 6, 2024
Atomic Allies , Expanding your Outreach Impact for Nuclear Science Week Webinar, <i>Panelist</i>	Sept. 13, 2023
American Nuclear Society , Social Media for Local & Student Sections Webinar, <i>Panelist</i>	Aug. 3, 2023
ANS Student Conference , Saturday Plenary, <i>Panelist</i>	Apr. 15, 2023
ANS Student Conference , Holding Onto a Professional/Personal Identity in Your Career, <i>Panelist</i>	Apr. 15, 2022
Women in Nuclear Region III , Academia Panel, <i>Panelist</i>	Aug. 19, 2021
Generation Atomic , Nuclear Advocacy Lab, <i>Seminar</i>	June 17, 2021
American Nuclear Society Trinity Local Section , <i>Keynote</i>	May 21, 2021
Los Alamos National Lab , Nuclear Engineering and Nonproliferation Division, <i>Seminar</i>	May 11, 2021
American Nuclear Society , What Does Earth Day Mean To You? Webinar, <i>Panelist</i>	Apr. 22, 2021
ANS Student , Interested in Nonproliferation? Opportunities to Build Your Expertise, <i>Panelist</i>	Apr. 10, 2021
ANS Student Conference , Taking Care of You, <i>Workshop</i>	Apr. 8, 2021
Purdue University , Women in Nuclear Student Section, <i>Seminar</i>	Mar. 3, 2021
American Nuclear Society , Graduate Fellowships, <i>Webinar</i>	Dec. 21, 2020
Stanford & Moscow Engineering Physics Institute , Young Professionals Nuclear Forum, <i>Speaker</i>	Dec. 9, 2020
NAYGN , Science Communication Series, <i>Speaker</i>	Oct. 13, 2020
UW-Madison , American Nuclear Society Student Section, <i>Seminar</i>	Oct. 8, 2020

Nuclear Energy Agency , Global Nuclear Science and Engineering Commencement, <i>Speaker</i>	Aug. 27, 2020
ANS Annual , Communicating Safety & Risk to the Public, <i>Panelist</i>	June 9, 2020
ANS Young Professionals Congress , Advocating to Policymakers, <i>Panelist</i>	Nov. 16, 2019
Ecomodernism , Nuclear Big Vs. Small, <i>Panelist</i>	Sept. 2019
Idaho National Laboratory , Atomic Week 2019, <i>Workshop</i>	Aug. 6, 2019
ANS Annual , From Nuclear Advocacy to Policy, From Local to Global, <i>Panelist</i>	June 13, 2019
ANS Annual , Results of the April 2019 "Fastest Path to Zero" Summit and Unconference, <i>Panelist</i>	June 12, 2019
ANS Winter , Advocacy and Communication: A Clean Energy Discussion, <i>Panelist</i>	Nov. 1, 2017
Nuclear Innovation Bootcamp , <i>Pitch Competition</i>	July 28, 2017
Environmental Progress , Atomic Vintage, <i>Lightning Talk</i>	July 26, 2017
Fermilab , QuarkNet Summer Research Program, <i>Symposium</i>	Aug. 2012

SHORT COURSES AND OTHER TRAINING

Nuclear Material Control and Accountability (MC&A) and Metallography Pilot Case Study	Knoxville, TN
NNSA Office of Defense Nuclear Security and Oak Ridge National Lab	Aug. 2023
International Nuclear Facilities Experience	AT/SK/CZ/DE/NL
Argonne National Lab and Texas A&M University	May 2023
IAEA Safeguards: From Theory to Practice	Richland, WA
Pacific Northwest National Lab	Mar. 2023
Machine Learning for International Safeguards	Virtual
NNSA Office of International Nuclear Safeguards (NA-241)	July 2022
Nuclear Engineering Student Delegation	Washington, D.C.
Independently student-organized. Delegation policy statement (link)	Sept. 2021
Young Professionals Nuclear Forum	Virtual
Stanford Center for International Security and Cooperation and the Moscow Engineering Physics Institute	Dec. 2020
International Safeguards Policy and Information Analysis-Intensive Summer Course	Virtual
Middlebury Institute of International Studies at Monterey and Lawrence Livermore National Lab	June 2021
Nuclear Security: The Nexus Between Policy and Technology	Virtual
University of California, Berkeley Course NE C285	Jan. 2021 - Apr. 2021
Nonproliferation and International Safeguards Summer Course	Virtual
Pacific Northwest National Lab	June 2020
Nondestructive Assay (NDA) Summer Course	Los Alamos, NM
Los Alamos National Lab	June 2018
Nuclear Innovation Bootcamp	Berkeley, CA
University of California, Berkeley	July 2017
Dr. G. Robert Keepin Nonproliferation Science Summer School	Los Alamos, NM
Los Alamos National Lab	Jun. - Aug. 2017
Geologic Storage of Nuclear Spent Fuel	Oskarshamn, Sweden
Swedish Royal Institute of Technology (KTH) & Swedish Nuclear Fuel and Waste Management Co. (SKB)	June 2015